**OBSERVATION CHECKLIST**

**ABERDARE TECHNICAL COLLEGE**

**INFORMATION COMMUNICATION TECHNOLOGY**

**MODULE 2, TERM 2**

| | |
|---|---|
| Unit | Computer Repair and Maintenance |
| Unit code | 0714 351 04A |
| Element 3 | Test Computer Component Functionality |
| Assessment | Practical test: Testing RAM using MemTest86 |

**Candidate and assessor details**

| | |
|---|---|
| Candidate name | |
| Registration no. | |
| Station / computer no. | |
| Date | |
| Assessor name | |
| Start time / End time | |

**SECTION A: CUTTING LIST (REQUIREMENTS)**

*Assessor to confirm that all items are available before the assessment begins.*

| No. | Item | Quantity | Provided by |
|---|---|---|---|
| 1 | Computer to be tested (desktop or laptop) with at least one RAM module installed | 1 | Institution |

| No. | Item | Quantity | Provided by |
|---|---|---|---|
| 2 | Second working computer with internet access (for downloading and preparing the USB) | 1 | Institution |
| 3 | Internet connection (for downloading the MemTest86 USB image) | 1 | Institution |
| 4 | USB flash drive of at least 1 GB (will be erased) | 1 | Candidate |
| 5 | MemTest86 USB image file (memtest86-usb.zip containing imageUSB.exe and memtest86-usb.img) | 1 | Candidate (downloaded during the test) |
| 6 | Rufus (optional alternative tool for writing the image) | 1 | Candidate (downloaded if needed) |
| 7 | Answer sheet | 1 | Institution |
| 8 | Pen | 1 | Candidate |

**SECTION B: PRACTICAL CHECKLIST (30 MARKS)**

*Award the full mark only when the candidate performs the step correctly without assistance. Award 0 if the step is not done or is done incorrectly.*

**Task 1: Preparation and download (5 marks)**

| No. | Observable performance criteria | Max marks | Marks awarded | Remarks |
|---|---|---|---|---|
| 1.1 | Opens a web browser and goes to the official MemTest86 site or direct download link | 1 | | |

| No. | Observable performance criteria | Max marks | Marks awarded | Remarks |
|---|---|---|---|---|
| 1.2 | Downloads the memtest86-usb.zip file successfully | 1 | | |
| 1.3 | Extracts the ZIP file using Extract All | 1 | | |
| 1.4 | Identifies imageUSB.exe and memtest86-usb.img in the extracted folder | 1 | | |
| 1.5 | Records the RAM information of the test computer (total size, speed, slots used) from Task Manager > Performance > Memory | 1 | | |
| | **Sub-total** | **5** | | |

**Task 2: Creating the bootable MemTest86 flash drive (6 marks)**

| No. | Observable performance criteria | Max marks | Marks awarded | Remarks |
|---|---|---|---|---|
| 2.1 | Backs up or checks the flash drive and inserts it into the working computer | 1 | | |
| 2.2 | Runs imageUSB.exe as Administrator | 1 | | |
| 2.3 | Selects only the correct flash drive in Step 1 (verifies drive letter and size) | 1 | | |
| 2.4 | Selects "Write image to UFD" in Step 2 | 1 | | |
| 2.5 | Selects memtest86-usb.img in Step 3 | 1 | | |

| No. | Observable performance criteria | Max marks | Marks awarded | Remarks |
|---|---|---|---|---|
| 2.6 | Clicks Write, confirms the warning, waits for completion, and safely ejects the flash drive | 1 | | |
| | **Sub-total** | **6** | | |

**Task 3: Booting from the flash drive (5 marks)**

| No. | Observable performance criteria | Max marks | Marks awarded | Remarks |
|---|---|---|---|---|
| 3.1 | Switches off the test computer and inserts the prepared flash drive | 1 | | |
| 3.2 | Presses the correct boot menu key at start-up | 1 | | |
| 3.3 | Selects the USB flash drive from the boot menu | 1 | | |
| 3.4 | Enters BIOS/UEFI and adjusts settings (Secure Boot / USB boot) only where needed, and records the change | 1 | | |
| 3.5 | MemTest86 loads and the main screen is displayed | 1 | | |
| | **Sub-total** | **5** | | |

**Task 4: Running the memory test (8 marks)**

| No. | Observable performance criteria | Max marks | Marks awarded | Remarks |
|---|---|---|---|---|
| 4.1 | Starts the test using the default settings (Enter / Start Test) | 1 | | |
| 4.2 | Records the RAM size detected by MemTest86 | 1 | | |
| 4.3 | Records the RAM type and speed (DDR3, DDR4, etc.) | 1 | | |
| 4.4 | Notes the test number, pass number, and elapsed time | 1 | | |
| 4.5 | Lets the test run without interrupting, touching, or powering off the computer | 1 | | |
| 4.6 | Allows at least 1 full pass to complete | 2 | | |
| 4.7 | Monitors the Errors counter and records the number of errors and the test during which they occurred | 1 | | |
| | **Sub-total** | **8** | | |

**Task 5: Results and report (6 marks)**

| No. | Observable performance criteria | Max marks | Marks awarded | Remarks |
|---|---|---|---|---|
| 5.1 | Reads and records the final result correctly (PASS or FAIL) | 1 | | |
| 5.2 | Exits MemTest86 correctly using Esc / Exit | 1 | | |

| No. | Observable performance criteria | Max marks | Marks awarded | Remarks |
|---|---|---|---|---|
| 5.3 | Removes the flash drive and restarts the computer normally | 1 | | |
| 5.4 | Locates and views the report file on the USB (EFI\BOOT folder) using a working computer | 1 | | |
| 5.5 | Fills in the answer sheet as the work progresses (not from memory afterwards) | 1 | | |
| 5.6 | Observes safety and procedure throughout (no unplugging or powering off during the test, no unauthorised BIOS changes) | 1 | | |
| | **Sub-total** | **6** | | |

**Practical checklist summary**

| Task | Max marks | Marks awarded |
|---|---|---|
| Task 1: Preparation and download | 5 | |
| Task 2: Creating the bootable flash drive | 6 | |
| Task 3: Booting from the flash drive | 5 | |
| Task 4: Running the memory test | 8 | |
| Task 5: Results and report | 6 | |
| **TOTAL PRACTICAL** | **30** | |

**SECTION C: PRODUCT CHECKLIST (5 MARKS)**

*Assess the finished products submitted by the candidate.*

| No. | Product criteria | Max marks | Marks awarded | Remarks |
|---|---|---|---|---|
| 1 | Bootable MemTest86 flash drive is functional and loads MemTest86 on the test computer | 1 | | |
| 2 | Answer sheet is fully completed (all 10 entries filled in) | 1 | | |
| 3 | RAM details recorded (size, type, speed) are accurate and match the computer tested | 1 | | |
| 4 | Test results recorded (passes, time, errors, PASS/FAIL) are consistent with the MemTest86 report | 1 | | |
| 5 | Conclusion on RAM health is logical and gives a correct recommendation (for example reseat the RAM, test one stick at a time, or replace the module if errors are found) | 1 | | |
| | **TOTAL PRODUCT** | **5** | | |

# Oral questions checklist(30 marks)

| No. | Oral question | Expected answer | Max marks | Marks awarded | remarks |
|---|---|---|---|---|---|
| 1 | What does POST stand for? | Power-On Self-Test. | 2 | | |

| | | | | | |
|---|---|---|---|---|---|
| 2 | Why does a computer run POST when it is switched on? | To check that the main hardware is present and working before the operating system loads. | 2 | | |
| 3 | Name two parts of the computer that POST checks. | Any two of: CPU, RAM, video card, keyboard, storage (hard drive/SSD), motherboard. | 2 | | |
| 4 | What does one short beep usually mean? | POST passed (completed successfully). | 2 | | |
| 5 | What does BIOS stand for, and what is UEFI? | Basic Input/Output System. UEFI is the modern replacement for BIOS. | 2 | | |
| 6 | What is the CMOS battery used for? | It keeps the BIOS settings and the clock (date and time) when the computer is switched off. | 2 | | |
| 7 | The computer has no beep, no display, and no fan. What is the likely problem? | A power problem (power supply, power cable, or power button). | 2 | | |
| 8 | Where can you find the meaning of a beep code? | In the motherboard or computer manual, or a beep code chart. | 2 | | |
| 9 | Name two tools used to test a computer that fails POST. | Any two of: POST card, multimeter, PSU tester, known-good parts, motherboard speaker. | 2 | | |
| 10 | Give two reasons why we test computer components. | Any two of: to find the faulty part, to save time and money, to confirm the repair worked, to | 2 | | |

| | | | | | |
|---|---|---|---|---|---|
| | | avoid the fault returning, for safety. | | | |
| 11 | What is swap testing? | Replacing a suspected faulty part with a part that is known to be good, to see if the fault goes away. | 2 | | |
| 12 | What is MemTest86 used for? | Testing the RAM (memory) for errors. | 2 | | |
| 13 | Why do you run MemTest86 from a USB flash drive and not inside Windows? | So Windows does not interfere with the test. It runs outside the operating system. | 2 | | |
| 14 | What should you do if MemTest86 shows errors? | The RAM is faulty. Test one RAM stick at a time to find the bad one, then replace it. | 2 | | |
| 15 | Name one software tool used to check the hard drive, and one used to check the CPU temperature. | Hard drive: CrystalDiskInfo. CPU temperature: Core Temp. | 2 | | |
| | **TOTAL** | | **30** | | |

**SUMMARY OF SCORES**

| Assessment | Max marks | Marks awarded |
|---|---|---|
| Practical checklist | 30 | |
| Product checklist | 5 | |
| Oral checklist | 30 | |
| | | |

**DECISION OUTCOME**

*Tick one.*

| | |
|---|---|
| ☐ Competent (C)<br><br>Below 49 | ☐ Not Yet Competent (NYC)<br><br>Above 50 |

**Reason for decision / areas needing improvement:**

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**ASSESSOR FEEDBACK**

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**CANDIDATE FEEDBACK**

*Candidate to comment on the assessment, and tick one:*

☐ I am satisfied with the assessment process and the decision.

☐ I am not satisfied and wish to appeal.

**Comments:**

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**SIGNATURES**

| | Name | Signature | Date |
|---|---|---|---|
| Assessor | | | |
| Candidate | | | |

**The end.**